

Dephasing Channel

$$|0\rangle_S \rightarrow \sqrt{1-p} |0\rangle_S \otimes |0\rangle_E + \sqrt{p} |0\rangle_S \otimes |2\rangle_E$$

$$|1\rangle_S \rightarrow \sqrt{1-p} |1\rangle_S \otimes |0\rangle_E + \sqrt{p} |1\rangle_S \otimes |2\rangle_E$$

→ Qubit Scatters with probability p

→ Channel picks preferential basis for system
↳ shown for $\{|0\rangle, |1\rangle\}$ above

→ Continuous Dephasing

$$\rho \rightarrow R_2(\theta) \rho R_2^\dagger(\theta) \quad \rightarrow \text{Follows Gaussian distribution of } \theta$$

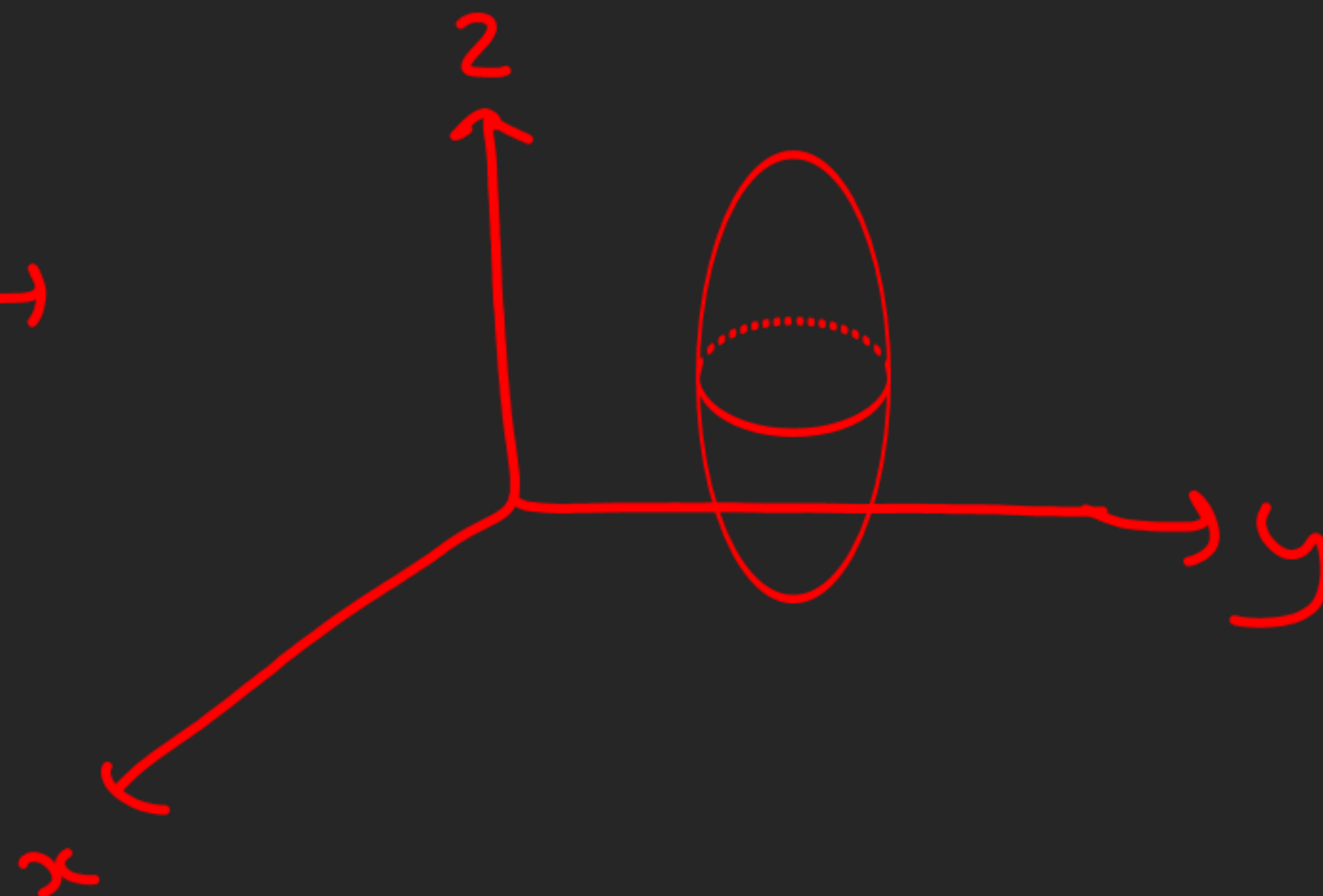
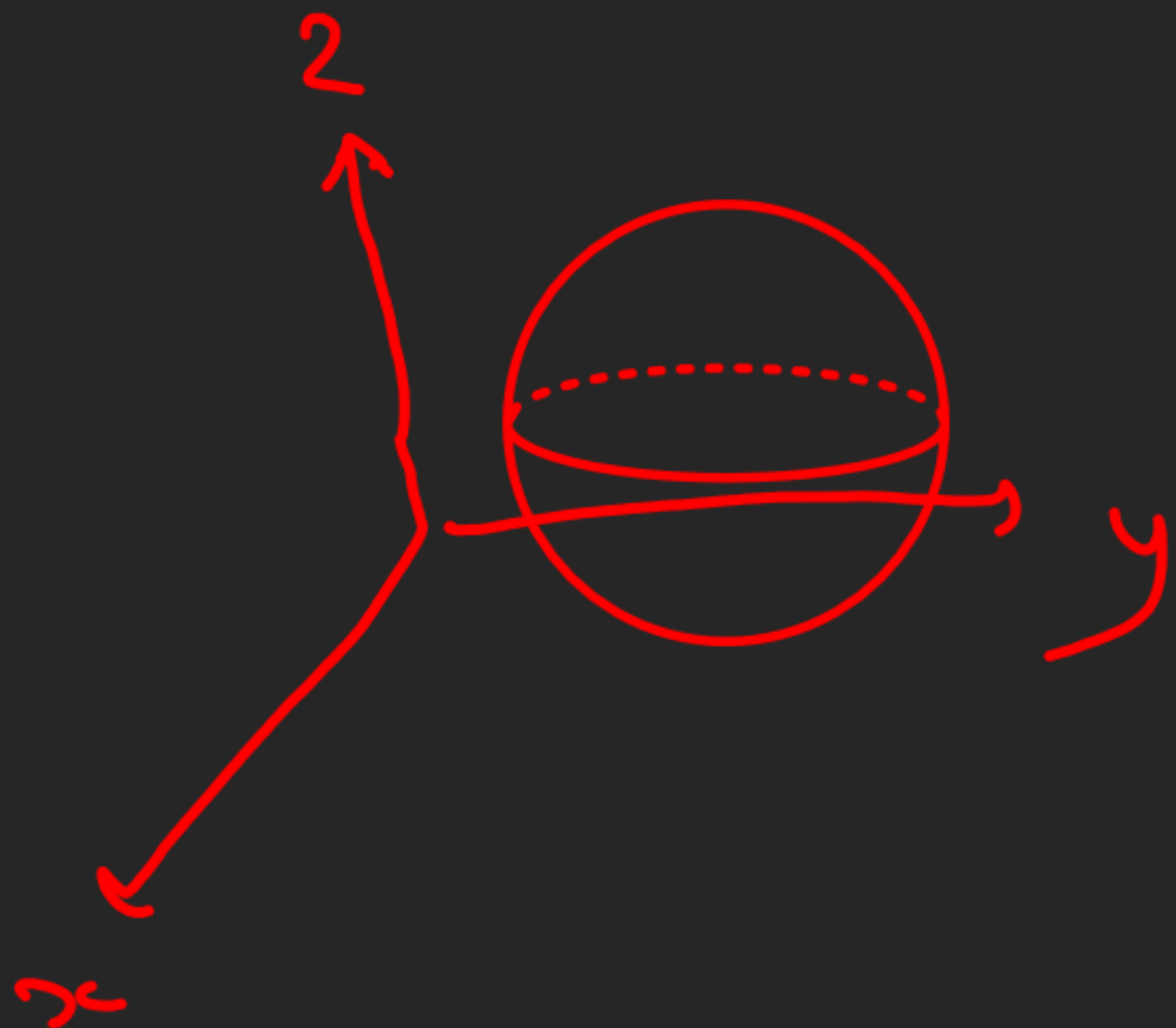
$R_2(\theta) \rightarrow$ Phase kick

$$R_2(\theta) = e^{-i\frac{\theta^2}{2}} \rightarrow \text{Rotation of Bloch vector along } z\text{-axis}$$

$$p(\theta) = \frac{1}{\sqrt{4\pi\lambda}} e^{-\theta^2/4\lambda} \quad \rightarrow \text{For } N(0, 2\lambda)$$

$$\rho(\theta) = R_2(\theta) \rho R_2^\dagger(\theta)$$

$$\rho' = \int_{-\infty}^{\infty} p(\theta) \rho(\theta) \cdot d\theta = \begin{bmatrix} \rho_{00} & \rho_{01} e^{-\lambda} \\ \rho_{10} e^{-\lambda} & \rho_{11} \end{bmatrix}$$



No Pancake Theorem

There exists no quantum channel which causes only 1 component to decay

